Assessed Coursework 3

Vulnerabilities Assessments and Remediation

Module Name	**Cyber Security Fundamentals**			
Coursework	**Vulnerabilities Assessments and Remediation**			
Report Deadline	**Time:**	**23:59**	**Dates:**	**9th Nov 2026**
ACW3 % Contribution to final module mark	**30%**		**Notional hours**	**3-4 hrs per week**
Solo or Group ✓	**Solo**	**x**	**Team**	**✓ (5-6 per team)**
Submission Instructions	**See details under 'Submission Requirements'**			
Who Will Mark This? ✓	**Instructors ✓**		**Postdoc**	**Other**
Team or Individual? ✓	**Team**		**Individual**	**Both ✓**
Please note: This coursework cannot be redone.				

Overview

Assessed Coursework 3 (ACW3) leverages what you have learnt to formulate and execute exploits against a targeted environment (e.g., system, web, operating system, network, IoT, and others) to discover vulnerabilities. The practicum has three components: **vulnerability identification, exploitation (pen-testing),** and **remediation**. The **outcomes and explanations are to be demonstrated as well as documented in a report** describing the vulnerabilities identified, exploits used, and the defence approaches that could be applied to **mitigate or remediate** the identified vulnerabilities. Mitigation reduces the impact of a threat when it cannot be eliminated, while remediation completely removes the threat.

ACW3 is **a team project**. You work in the **same team as in ACW1**. You should self-organise your team to try to optimize efficiency, as different members may have strengths in different areas/aspects of the work. The **pen-testing stage** needs to be demonstrated to the instructors during assessment, and it will **contribute a maximum of 20%** of your overall module marks. The **report will contribute a maximum of 10%** of your overall module marks.

Building the Targeted Environment

You MUST build your own target environments for this coursework on your own laptops natively or with a virtual machine (VM). DO NOT FIND any online or operational environment in the SIT network or the Internet for your target.

NOTE - If you use the online target environment and it has been flagged, you will take full responsibility for any consequences.

To build the environment, you can refer to any online guide, or you can refer to the official CVE (Common Vulnerabilities and Exploitation) website to know the actual vulnerability. There are a lot of vulnerabilities on the website, but please select ONLY vulnerabilities that are listed in the separate table below, with any level of severity for your coursework. You may also propose your own CVE, but subject to APPROVAL by the instructors (please send an email before you start working on it)

The list of the CVEs can be found in the xSite in the INF2005-ACW3-CVE-List section.

There is no minimum number of targeted environments. You can combine multiple vulnerabilities in a single targeted environment but make sure that the environments are **ready for demonstration** by the due date (**AY2026T1 Week 11**, see **ACW3 Demo** section for demo schedule details).

Pen-testing

In this stage, you must bring up your target environment and pen-testing environment (e.g., Kali Linux or others) before the demonstration; you ONLY need to show step-by-step how to **identify and exploit** the vulnerabilities in the targeted environment. The remediation steps are OPTIONAL to demonstrate.

NOTE - You may use any hacking technique/tool (including AI-assisted tools) to complete this coursework.

Report

- The report should **not exceed 10 pages (excluding the cover, references, and appendix),** use **11-point Calibri,** and should be submitted **in PDF format**.
- The report name format: **Report_P<labgroup_number>-<team_number>.pdf** (e.g., **Report_P2-11.pdf),** and list team members' **full names** and **student IDs** on the **first page.**
- The report should include the following points:
  - Summary of **the vulnerability exploits and remediation** that your team conducted in the pen-testing stage in a table as shown below on the **second page**.
  - The number of vulnerabilities is **unlimited**, but **each group** needs to accumulate a total severity score (based on the CVSS) of **at least 20 points**. The scale is **Critical (9.0–10.0), High (7.0–8.9), Medium (4.0–6.9), Low (0.1–3.9), and None (0.0).**
  - For each vulnerability documented, describe the exploit technique/tool used, add the screenshots **in the appendix** for conciseness, and describe suitable defence mechanism(s).

#	Severity Scale	Vulnerability	Used Exploit	Recommended Remediation
1	6	SQLI	Script A	Patch or Fix A
2	5	Broken Auth & Session Mgmt	Code URL	Implement Auth

- Each team needs to **submit only one (1) report** to the Xsite Dropbox by **Monday, 23:59, 9th Nov 2026**, with **one (1) declaration of originality** signed by **all team members**.

Note: It is allowed to use the generated AI tools for the report, but please keep in mind the SIT policy on using GenAI for exam/assessment, and clearly state which part was (partially) generated by the GenAI tools.

Marks Allocation

The marks for ACW3 are allocated in two parts as follows:

1. **Pen-testing demo (20 marks)** – All team members must be presenting the work, and the scoring rubrics are as follows:
   a. Demonstrate the exploitation of three **(3) vulnerabilities** approved by the instructor.
   b. Ability to answer question(s) posed by the instructor.
   c. The reward for the final marks will be based on the **highest level of severity** (**Critical or High** is rewarded with **100%** of total marks, **Medium** is rewarded with **95%** of total marks, **Low or None** is rewarded with **90%** of total marks).
2. **Report (10 marks)** – Submitted by one team representative before the deadline. **A late submission is accepted**, but it will be given a **5% penalty.**

Peer Review

A team member's grade may be adjusted by deltas, if necessary, to represent fair recognition of individual contributions. A team member may download the ACW3 Peer Review Form (from Xsite), complete and submit it if that member feels there's **an imbalance of contribution** within the team. If there is no submission from a team, it will be assumed that **all members contributed equally**. Submit peer review forms to these instructors: ict_francis.mahendran@singaporetech.edu.sg, AF_Ng.Wei.Kai.Bennet@singaporetech.edu.sg, khengleong.tan@singaporetech.edu.sg, and ariscahyadi.risdianto@singaporetech.edu.sg, but you MUST **CC all members** for transparency by **Friday, 14th Nov 2025**.

Demo

Demo Schedule:

Date	Time Slot	Teams	Venue
Tuesday, 10th Nov 2026	9:00am – 11:00am	Group P5 1-7	E2-03-02-SR220
	11:00am – 1:00pm	Group P6 1-12	
Thursday, 12th Nov 2026	9:00am – 11:00am	Group P4 1-10	E2-05-05
	11:00am – 1:00pm	Group P2 1-12	
	2:00pm – 4:00pm	Group P3 1-9	
	4:00pm – 6:00pm	Group P1 1-14	

Demo Guidelines:

- Demo eval session for each team will be about **30 minutes**, including **10 minutes** of Q&A.
- **All members** must be present, and **each member** will be given a chance to demo or answer questions about their pen-testing results and analysis.
- Due to a time constraint, each vulnerability demo and/or question response must be made in one attempt. **There is no time for repeated, delayed, or prolonged attempts.**
- Teams MAY swap their demo dates from Tuesday to Thursday or vice versa if they can FIND another team that has **agreed** to swap, and they need to **inform** the instructors via email.
- If any team member is absent **without a valid reason** (e.g., can't finish the assignment, missed the bus, woke up late, …), a **0 (zero) mark** will be awarded for this component.
- If any team member is absent **with a valid reason** (e.g., MC), a separate demo session will be conducted for that member.
- All time slots are allocated **strictly for demo and questions** and NOT for setting up or pre-demo testing/debugging/familiarization.